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PALEOMAGNETISM OF MESOZOIC RED CHERT FROM CEDROS ISLAND AND THE SAN BENITO ISLANDS, BAJA CALIFORNIA, MEXICO REVISITED

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Abstract. Previous paleomagnetic study of Mesozoic subduction-complex rocks on Cedros Island and the San Benito Islands, Mexico, shows that these rocks have been remagnetized; a single-polarity magnetization was found in chert sections and underlying pillow basalt deposited during a mixed-polarity interval of the geomagnetic field. Reanalysis of the Cedros and San Benito chert samples (101 total) shows that 13 samples also retain a high blocking-temperature component of magnetization (600° to 680°C) indicating a polarity stratigraphy and deposition of the bedded chert near 2° ± 3° paleolatitude. These data are similar to those for red chert in California and Japan implying that many of the Mesozoic ophiolitic chert sequences now exposed around the Pacific rim were initially deposited within the equatorial zone of high biologic productivity. These new data also support accretion of the Cedros and San Benito chert to the American margin at tropical paleolatitudes requiring significant northward translation (and clockwise rotation) to bring them to their present position with respect to the continental interior.

Introduction

In a previous investigation of Mesozoic red radiolarian chert of Cedros Island and the San Benito Islands, Mexico (Figure 1), *Hagstrum and Sedlock* [1990] concluded that these oceanic rocks were magnetically overprinted during mid-Cretaceous subduction-related alteration [*Sedlock*, 1988a; *Baldwin and Harrison*, 1989]. A characteristic magnetization of consistent polarity and direction was observed within sections of chert and underlying pillow basalt deposited, between Late Triassic and Early Cretaceous time [*Sedlock and Isozaki*, 1990], during a mixed-polarity interval of the geomagnetic field [e.g., *Harland et al.*, 1982]. In addition, failure of a small-scale intrastratal fold test at one site and significantly decreased dispersion of the paleomagnetic data after application of larger-scale structural corrections, imply that remagnetization occurred after deposition of the chert sequence but prior to dismemberment of the oceanic protolith [see also *Hagstrum and Sedlock*, 1991].

These results are nearly identical to those reported by *Shibuya and Sasajima* [1986] for samples of Mesozoic red chert from the Mino terrane in central Japan: a characteristic magnetization of uniform polarity is observed in all samples that fails an intrastratal fold test, but passes a large-scale tectonic fold test. *Shibuya and Sasajima* [1986] inferred that this magnetization was acquired as a low-temperature chemical remanent magnetization between intrastratal folding and tectonic folding of the chert sequence during accretion of the Mino terrane to the Asian landmass. Similarly, a uniform-polarity magnetization of consistent direction was observed in samples from Lower and Middle Jurassic red chert, underlying pillow lavas, and overlying clastic sedimentary rocks of the Marin Headlands terrane in northern California [*Curry et al.*, 1984]. This magnetization apparently predates deformation of the siltstone layers and is consistent with a large post-subduction clockwise rotation of the Marin Headlands block (>90°) proposed to explain the anomalous orientation of the structural data [*Warhaftig*, 1984]. Thus, these and other accreted oceanic rocks of the Franciscan Complex were likely remagnetized prior to structural deformation by low-temperature chemical alteration during subduction and accretion at the American continental margin [*Hagstrum*, 1990].

At one chert locality (out of 9) in the Mino terrane, however, *Shibuya and Sasajima* [1986] discovered a high blocking-temperature component of magnetization that showed stratigraphically-related polarity intervals. This component of magnetization passed both intrastratal and tectonic fold tests and was interpreted by *Shibuya and Sasajima* [1986] to have been acquired shortly after deposition of the chert sequence near a paleolatitude of 1° ± 3°. Recently, *Hagstrum and Murchey* [1991] found a high blocking-temperature component of magnetization showing a polarity stratigraphy in samples from two sequences of Lower Jurassic red chert in the Franciscan Complex of northern California. Although strongly overprinted in all samples by a magnetization of uniform polarity, this higher-temperature component also indicates equatorial paleolatitudes consistent with deposition of the bedded chert within the equatorial zone of high biologic productivity [*Murchey*, 1984].

Recognition that red chert from California and Japan is likely to intermittently retain a potentially primary component of magnetization at the highest blocking-temperature levels sent us back to reexamine the demagnetization data for the Cedros and San Benito chert samples. Although pilot thermal demagnetizations for the collection (1 or 2 samples per site) had reached 675°C (Curie temperature for hematite), most samples were not heated above 620°C. Some samples showed directional changes at higher temperatures and low remanent intensities that were originally interpreted as noise. For this study, continued thermal demagnetization treatments were undertaken on those samples from the Cedros and San Benito collection that might contain an additional high-temperature component of magnetization. In this report we present the results of this reexamination. The presence of a distinquishable high-temperature component showing normal and reversed polarities is documented for 13 samples out of the 101-sample collection. This high-temperature component also indicates deposition of the Cedros and San Benito chert beds at equatorial paleolatitudes on the ancient sea floor.

Geologic Background

Mesozoic oceanic rocks in the west-central region of the Baja California peninsula are assigned by *Sedlock* [1988a, 1988b] to three shallowly-dipping tectonic units: (1) an upper plate consisting of Triassic to Lower Cretaceous arc-related rocks and mid-Cretaceous turbidite, (2) a lower plate consisting of regionally-metamorphosed subduction complex rocks, and (3) a fault-bounded serpentinite-matrix melange separating the lower and upper plates (Figure 1). The lower-plate subduction complex rocks (Western Baja terrane) are similar in many respects to those of the Franciscan Complex in northern California, and have been subdivided into three distinct

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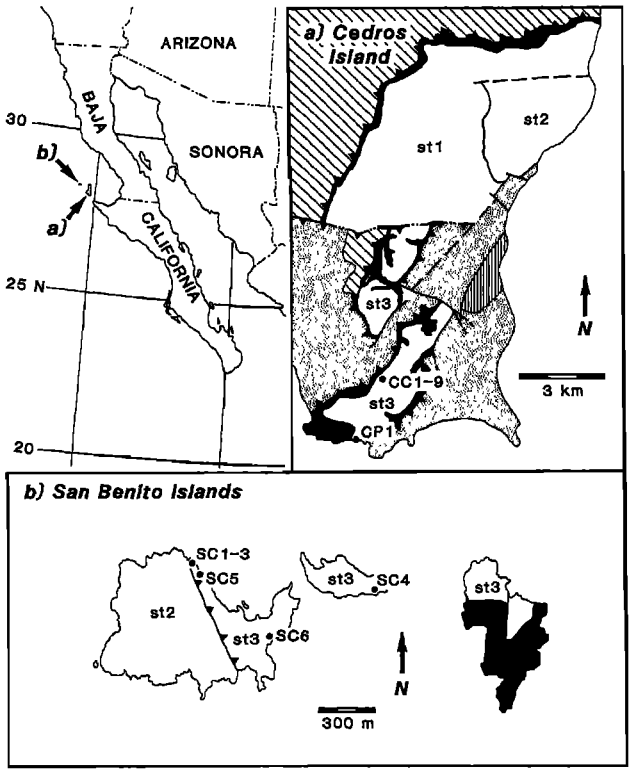


Fig. 1. Location map of Cedros Island and the San Benito Islands off the western coast of Baja California, Mexico. Geologic maps (a) of southeastern Cedros Island and (b) of the San Benito Islands, show paleomagnetic sampling sites; lower-plate subterranes st1, st2, and st3 of the Western Baja terrane (no pattern); fault-bounded serpentinite-matrix melange (black); upper-plate rocks of the Choyal terrane (diagonal lines); overlying Upper Cretaceous Valle Formation (stippled pattern); and Miocene and Pliocene units (vertical lines). Heavy lines indicate fault contacts (teeth on upper plate); lighter lines indicate depositional contacts.

subterranes based on variations in metamorphic grade and in intensity of structural disruption [*Sedlock*, 1988b]. The chert samples studied by *Hagstrum and Sedlock* [1990], and reexamined in this study, are from fault-bounded blocks within the least metamorphosed and least deformed subterrane (subterrane 3) of the Western Baja terrane.

Deformation elements in subterrane 3 include a weakly-developed spaced foliation subparallel to bedding, minor open-to-tight folds of this foliation, thrust faults that formed during and after folding, and normal faults. The foliation, folds, and thrust faults most likely formed during incipient blueschist-facies metamorphism and accretion of the oceanic rocks to the American margin [*Sedlock*, 1988a, 1988b]. $^{40}Ar/^{39}Ar$ ages for higher-grade rocks of the Western Baja terrane indicate that peak blueschist-facies metamorphism occurred at about 110 Ma, and uplift ages for subterrane 3 are as old as 75 Ma [*Baldwin and Harrison*, 1989]. Normal faults are interpreted to have formed during uplift, and faults of all types bound the innumerable blocks within the Western Baja terrane [*Sedlock*, 1988b].

The reconstructed stratigraphic sequence of subterrane 3 in ascending order is ocean-floor basalt, radiolarian ribbon chert, and thin-bedded siliciclastic turbidite. Original thicknesses cannot be accurately determined due to difficulties in correlating units between individual blocks. The basal sequence of basalt is at least several hundred meters thick and includes pillow basalt, massive flows, volcaniclastic breccia, and thin interbeds of limestone and chert. The thickest sections of overlying chert are 40 m thick on Cedros Island and 30 m thick on San Benito West Island. Radiolarian biostratigraphy indicates that the chert sequence ranges in age from Late Triassic to late Early Cretaceous [*Sedlock and Isozaki*, 1990]. Overlying the chert is a sequence of siliciclastic turbidite that is at least 250 m thick. Thus, the protolith of the subduction complex rocks is interpreted as Late Triassic oceanic crust that was first covered by chert in the open ocean and later by siliciclastic turbidite as the oceanic plate neared the continental margin in late Early Cretaceous time [*Sedlock*, 1988b].

Paleomagnetism

Sections of red chert in 5 fault-bounded blocks were sampled by *Hagstrum and Sedlock* [1990] on Cedros Island and the San Benito Islands at 15 sites. An additional site was sampled in pillow basalt beneath the northeast-trending Punta Prieta ridge on Cedros Island (CP1). Between 5 and 8 samples were collected primarily as hand samples at each site, and multiple sites were collected in the largest two sections of chert on Cedros Island (CC1-9) and on San Benito West Island (SC1-3; Figure 1). Attitudes of the chert beds are generally consistent within each block with the exception of the Punta Prieta block on Cedros Island. Most of the sites on Cedros Island are in a 40-m-thick homoclinal section of chert, but other sites are in more complex portions of this exposure with varying attitudes. The deformation represented by this variation is predominantly due to intrastratal folding and larger scale soft-sediment deformation; a number of faults with minor displacements also cut the block. Because the uncorrected direction of the overprint magnetization is consistent in most samples of chert from the Punta Prieta block, *Hagstrum and Sedlock* [1990] applied a single correction to all sites based on the regional attitude of beds within the block. In this study,

TABLE 1. Primary Directions and Poles for Mesozoic Chert from Cedros Island and the San Benito Islands

Sample	Uncorrected I	Uncorrected D	Strike/ Dip	Corrected I	Corrected D	Pole Lat	Pole Long
Cedros Island (28.1°N, 244.8°E)							
CC2-1*	-24.2	16.8	150/38	5.0	21.2	57.4	22.7
CC6-1	-11.1	22.6	165/30	7.6	23.3	57.1	18.2
-2*	29.1	200.0	135/20	10.7	202.9	49.9	27.8
-3	33.3	162.6	130/20	21.2	171.0	50.0	78.6
-4	-16.8	54.9	130/20	2.6	54.3	31.7	172.2
CC8-1*	12.2	196.0	92/07	5.4	195.7	55.7	36.1
-2	-27.4	42.8	92/07	-22.0	40.7	34.2	14.2
CC9-1	41.2	208.2	145/40	4.2	215.1	44.7	10.8
-2	36.3	209.4	145/40	-0.5	214.7	46.7	8.8
-3	44.4	222.5	145/40	5.0	226.1	36.2	1.7
San Benito Islands (28.3°N, 244.4°E)							
SC2-1	47.6	165.3	347/74	10.6	120.3	23.5	134.0
SC4-1	75.6	125.3	356/70	8.7	95.2	2.5	148.1
-2	66.4	77.7	356/70	-3.4	82.7	5.6	159.4
SC5-1	-24.9	30.4	157/36	5.0	34.1	48.5	6.7
-2	-36.1	65.4	157/36	-0.1	65.7	21.2	346.5
SC6-1	65.5	206.2	332/99	10.9	76.3	14.7	336.0
Averages (inclinations only)							
All samples (N = 16):							
	36.8 (k = 9; α_{95}= 10.7°)						
				4.7 (k = 41; α_{95}= 4.8°)			

I and *D:* inclination and declination of paleomagnetic directions, uncorrected/corrected for tilt, in degrees; dip of chert beds 90° clockwise of strike; *Lat, Long:* N latitude and E longitude of virtual geomagnetic pole corrected for tilt, in degrees; N, number of entries averaged; k, precision parameter; α_{95}, 95% confidence limits.

*Group *B* samples.

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however, attitudes of the chert beds at individual sites are used as structural corrections (Table 1).

Initially, thermal demagnetization of 1 or 2 pilot samples from each site in the red chert was continued until the samples lost most (90-100%) of their remanent magnetic intensity [*Hagstrum and Sedlock*, 1990]. Maximum blocking temperatures are generally below 620° to 650°C, but in some cases are as high as the Curie temperature for hematite (675°C). Isothermal remanent magnetization experiments also indicate that hematite is the dominant magnetic mineral in these rocks [*Hagstrum and Sedlock*, 1990]. In almost all of these pilot samples a single characteristic component of magnetization was observed, so heatings of the remaining chert samples were done mostly at temperatures below 620° to 650°C. In some samples, the vector endpoints did not decay directly to the origin, indicating the presence of very weak high-temperature components. These components appeared to have random directions and were interpreted by *Hagstrum and Sedlock* [1990] as having no geologic significance. Other studies of red chert [*Shibuya and Sasajima*, 1986; *Hagstrum and Murchey*, 1991], however, have shown that very weak high-temperature components represent an early, likely primary component of magnetization in these rocks.

In this restudy, all chert samples of the Cedros and San Benito collection that had not been completely demagnetized and that showed any possibility of containing a second high-temperature component of magnetization were progressively demagnetized to temperatures exceeding 675°C. A consistent component was found in 13 samples at temperatures between about 600° and 680°C; in other samples this high-temperature component was completely overprinted to varying degrees by a secondary component of magnetization. Although a continuum of overprinting of the high-temperature component is present in the chert samples, for descriptive purposes we have defined 4 basic groups of behavior and show representative examples of each in Figure 2. In group *A* samples (13 samples, Figure 2a), the overprint magnetization is completely removed and a distinct high-temperature component is isolated; in group *B* samples (3 samples, Figure 2b), the overprint magnetization is removed and an endpoint is reached near the primary direction for other samples from the same site, but the high-temperature direction cannot be defined by least-squares lines; in group *C* samples (21 samples, Figure 2c), the overprint component is not completely removed and the final demagnetization direction is somewhere in between the two directions; and in group *D* samples (64 samples, Figure 2d), the high-temperature component has been completely obliterated by the overprint component. Remagnetization circle analysis was applied to group *B* and *C* samples, but intensities of the primary component were too weak and the remagnetization circles intersected only at the overprint direction.

Directions of the high-temperature component for samples from groups *A* and *B* are plotted in Figure 3 both before and after application of the structural corrections (Table 1). The directions show normal and reversed polarities within the Punta Prieta block on Cedros Island and have shallow inclinations (mean = 5° ± 5°) indicating near equatorial paleolatitudes (mean = 2° ± 3° north or south) after correction. Declinations for the Cedros and San Benito chert samples most likely have been dispersed by within-block and between-block vertical-axis rotations. Inclination values are significantly better grouped after correction ($N = 16$; $k_1/k_2 = 4.56$) indicating that the high-temperature magnetization most likely predates most, if not all, deformation of the chert beds.

Discussion and Conclusions

The presence of a high-temperature component of magnetization in a minority of the Cedros and San Benito chert samples confirms the conclusion of *Hagstrum and Sedlock* [1990] that these rocks were extensively remagnetized, probably during subduction-related alteration and accretion. There appears to be no stratigraphic relationship of remagnetization and preservation of the primary magnetization: a wide range of overprinting is observed within most sites. The presence of normal and reversed polarity directions in the Punta Prieta block and a positive "fold" test for the high-temperature component implies that it was acquired by the chert beds during or soon after deposition. These results are similar to those obtained from other chert sequences in northern California and Japan [*Hagstrum and Murchey*, 1991; *Shibuya and Sasajima*, 1986], indicating that many of the sequences of ophiolitic chert now exposed along the Pacific rim may have originally been deposited on the sea floor beneath the equatorial zone of high biologic productivity.

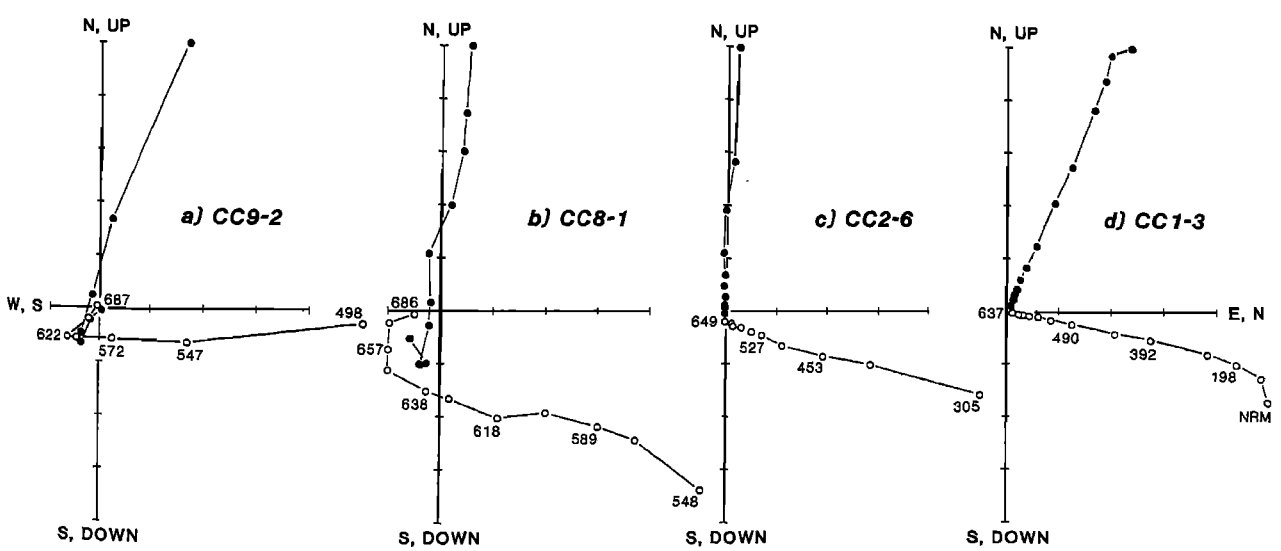


Fig. 2. Orthogonal vector plots of thermal demagnetization data for red chert samples showing range of overprinting of the primary component of magnetization. All directions *in situ* prior to application of bedding tilt corrections. Group *A* behavior (a): primary component preserved, axis ticks at 4.5 X 10^{-4} A m^{-1}; group *B* behavior (b): overprint component removed, axis ticks at 6.6 X 10^{-5} A m^{-1}; group *C* behavior (c): overprint component not removed, axis ticks at 5.0 X 10^{-4} A m^{-1}; (d) group *D* behavior: primary component obliterated, axis ticks at 2.1 X 10^{-3} A m^{-1}. Filled circles represent projection onto the horizontal plane, and unfilled circles onto the north-south vertical plane. Numbers indicate level of thermal demagnetization treatment in °C.

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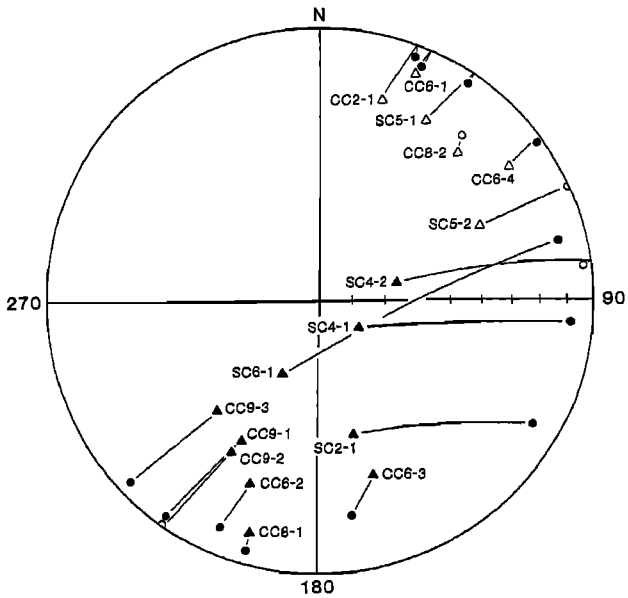


Fig. 3. Equal-area plot of primary high-temperature directions for red chert samples (Table 1) both before (triangles) and after correction for structural tilt (circles). Filled symbols indicate projection from the lower hemisphere and unfilled symbols from the upper hemisphere.

Deposition of the Cedros and San Benito chert within the equatorial productivity zone (≤25° north or south latitude) just prior to deposition of the overlying continentally-derived turbidite indicates that the chert was accreted to the American margin at tropical paleolatitudes, consistent with remagnetization of the chert near 12° north paleolatitude as inferred from structural arguments by *Hagstrum and Sedlock* [1990, 1991]. This is also consistent with the proposal of *Murchey* [1984] that the Marin Headlands chert of California followed a low-latitude trajectory near the equator during its depositional history and therefore was accreted far south of its present position with respect to North America. Geologic evidence and normal-polarity overprints in many oceanic rocks exposed along the western margin of North America indicate that these rocks were probably remagnetized and accreted during the Cretaceous Normal Polarity Superchron (~120-85 Ma). Declinations of the primary component for the chert (Figure 3) generally indicate vertical-axis rotations, averaging about 70° clockwise (assuming the northeast declinations represent normal polarity directions), relative to the continental interior. Northward translation and clockwise rotation of the Western Baja terrane during Late Cretaceous to early Tertiary time are consistent with plate reconstruction models [*Engebretson et al.*, 1985] and the interpretation of paleomagnetic data for many other terranes along the western American margin [*Beck*, 1980].

In summary, the paleomagnetic data for the Cedros and San Benito chert indicate that the chert was deposited near equatorial latitudes as the oceanic plate moved west to east beneath the biologic productivity high. The cherts were remagnetized before major deformation and disruption of the chert sequence at low paleolatitudes [*Hagstrum and Sedlock*, 1990, 1991] during the Cretaceous Normal Polarity Superchron, and were subsequently transported northward and rotated clockwise by arc-parallel strike-slip faulting associated with oblique subduction of oceanic plates along the American continental margin [*Engebretson et al.*, 1985] during mid-Cretaceous to middle Tertiary time.

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